

KABIR THAYANI

AI/ML Engineer

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in Computer Vision, Edge AI, and applied Deep Learning. Proven experience compressing and deploying production-grade models through Knowledge Distillation, achieving up to 25x model size reduction without accuracy loss. Strong background in building end-to-end ML systems using PyTorch, FastAPI, and Docker, with hands-on exposure to RAG pipelines and real-world AI deployment under resource constraints.

TECHNICAL SKILLS

Machine Learning & Vision: PyTorch, TensorFlow, CNNs, YOLOv8, OpenCV, Transfer Learning, Knowledge Distillation, Edge AI
Generative AI & NLP: RAG, LangChain, LLMs (LLaMA, Mistral, Gemma), Prompt Engineering, ChromaDB
Backend & MLOps: FastAPI, REST APIs, Docker, Git, ONNX, n8n Automation
Programming & Data: Python, SQL (PostgreSQL), NumPy, Pandas, React.js

PROFESSIONAL EXPERIENCE

Aarya Softex LLP

Python Developer Intern

Junagadh, India

June 2025 – August 2025

- Deployed scalable ML microservices using FastAPI with secure authentication and role-based access control, enabling concurrent production inference.
- Designed and implemented a Retrieval-Augmented Generation chatbot using LangChain and vector embeddings, reducing internal information retrieval time by 40%.
- Built a real-time OCR and NLP pipeline to extract structured data from unstructured PDFs, improving lead processing efficiency by 70%.
- Developed a low-latency voice AI agent integrating speech-to-text and text-to-speech pipelines for automated outbound workflows.
- Automated multi-step business processes using n8n, eliminating manual data entry and improving operational reliability.

KEY PROJECTS

NanoRay - Edge AI for Medical Imaging | PyTorch, Knowledge Distillation

- Compressed a 90MB ResNet50 teacher into a 3.6MB MobileNetV3 student for edge deployment.
- Achieved 25x model compression while maintaining diagnostic performance (84.15% accuracy), surpassing the teacher.
- Enabled offline, CPU-only inference for low-power medical devices.

CivicLens - AI-Based Road Damage Detection | YOLOv8, Computer Vision

- Built a real-time object detection pipeline to identify potholes and road hazards from video streams.
- Integrated GPS tagging and duplicate-report filtering to streamline civic issue reporting.

AI-Powered Applicant Tracking System | RAG, FastAPI, LLMs

- Implemented semantic resume search using ChromaDB and vector embeddings.
- Built an LLM-based resume parser converting PDFs into structured JSON, reducing screening time by 40%.

Syllaby - AI Study Planner | React, FastAPI, PostgreSQL, Ollama

- Developed an AI-driven study planner generating adaptive schedules, flashcards, and dashboards.
- Integrated local LLMs (Gemma 1.2B) for privacy-focused, offline content generation.

PUBLICATIONS

NanoRay V2: Bridging the Gap Between Transformers and Edge AI via Cross-Architecture Distillation
International Journal of Engineering Research & Technology (IJERT), Vol 15, Issue 01, Jan 2026. ISSN: 2278-0181

EDUCATION

Noble University

Bachelor of Technology (B.Tech) in Computer Science Engineering

- CGPA: 9.16/10

Junagadh, India

Expected 2026

CERTIFICATIONS

Foundation Course on IR4.0 Technologies - Microsoft, SAP, Edunet (2025) | **Intro to AI & ML** - Simplilearn (2025)